

5. There are a number of factors that should be taken into consideration when deciding what makes the 'best fuel'.

Information was collected about various factors for three fuels, **A**, **B** and **C**.

**Fuel A**

- Existing supplies will last around 50 years
- Releases 2.8 kJ of energy per gram of fuel burned
- Costs 0.03p per gram of fuel burned
- Burns very easily and no storage issues
- Releases carbon dioxide and water vapour when it burns

**Fuel B**

- There are infinite supplies of this fuel
- Releases 44.1 kJ of energy per gram of fuel burned
- Costs 0.18p per gram of fuel burned
- Burns very easily but can be difficult to store
- Releases water vapour when it burns

**Fuel C**

- Existing supplies will last around 250 years
- Releases 1.2 kJ of energy per gram of fuel burned
- Costs 0.04p per gram of fuel burned
- Burns very easily and fairly easy to store
- Releases carbon dioxide, sulfur dioxide and water vapour when it burns

This information was analysed by a group of students to decide what they considered to be the 'best fuel'.



- (a) Give the reason why the information about how easily each fuel burns was not useful to the students when reaching their decision. [1]

.....

.....

- (b) One of the students based his decision purely on a judgement of the available supply of each fuel. Choose the order that shows his conclusion. Place a tick (✓) in the appropriate box. [1]

|                                                                                   |                          |                          |                          |                          |                          |                          |
|-----------------------------------------------------------------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|
| best fuel                                                                         | <b>A</b>                 | <b>A</b>                 | <b>B</b>                 | <b>B</b>                 | <b>C</b>                 | <b>C</b>                 |
|                                                                                   | <b>B</b>                 | <b>C</b>                 | <b>A</b>                 | <b>C</b>                 | <b>A</b>                 | <b>B</b>                 |
|                                                                                   | <b>C</b>                 | <b>B</b>                 | <b>C</b>                 | <b>A</b>                 | <b>B</b>                 | <b>A</b>                 |
|  | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |

- (c) Which of the following statements best describes how the fuels affect the environment when they burn? Tick (✓) the correct answer and give the reason for your choice. [2]

all of the fuels contribute to acid rain and global warming when they burn

☐

fuels **A** and **C** contribute to acid rain and global warming when they burn

☐

only fuel **C** contributes to acid rain and global warming when it burns

☐

none of the fuels contribute to acid rain and global warming when they burn

☐

Reason .....

.....



(d) The cost efficiency of fuel **A** can be calculated as follows:

cost efficiency =  $\frac{2.8}{0.03} = 93.3 \text{ kJ/p}$

Use the information given for fuel **B** and this example to calculate the cost efficiency of fuel **B**. [2]

Cost efficiency = ..... kJ/p

(e) The students eventually agreed on the following rank order for the fuels.



In the table below, tick (✓) **all** the statements that are **correct** and could therefore have been used in deciding upon this order. [2]

|                                                                     | (✓) |
|---------------------------------------------------------------------|-----|
| fuel <b>C</b> will run out after fuels <b>A</b> and <b>B</b>        |     |
| fuel <b>C</b> is easier to store than fuel <b>A</b>                 |     |
| fuel <b>A</b> burns more easily than fuel <b>C</b>                  |     |
| fuel <b>B</b> is the cleanest fuel                                  |     |
| fuel <b>B</b> is easier to store than fuel <b>C</b>                 |     |
| fuel <b>B</b> will never run out                                    |     |
| fuel <b>A</b> is less harmful to the environment than fuel <b>C</b> |     |
| fuel <b>A</b> is less cost efficient than fuel <b>B</b>             |     |

